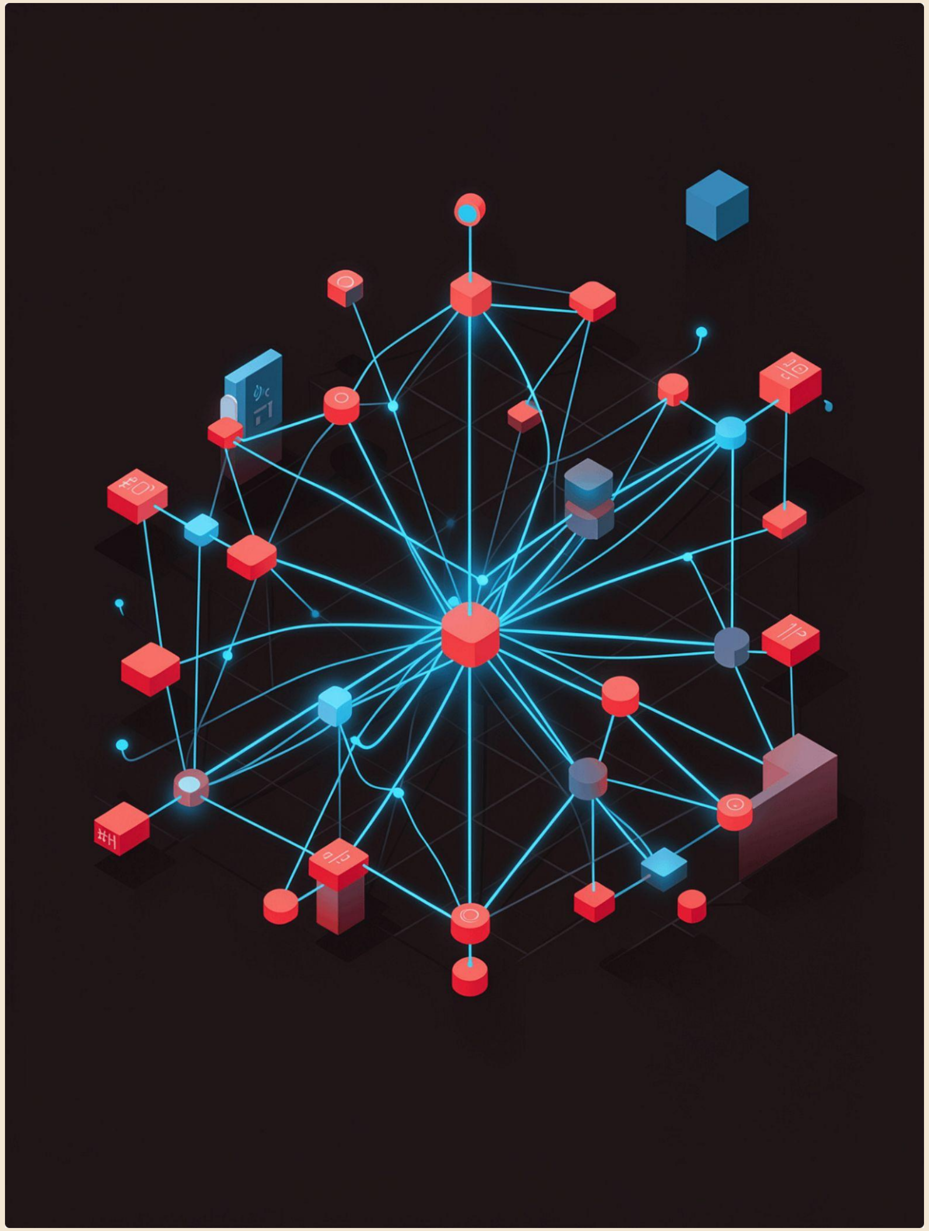
Regression in Machine Learning

Understanding Predictive Modeling using Regression



What is Regression?

Definition

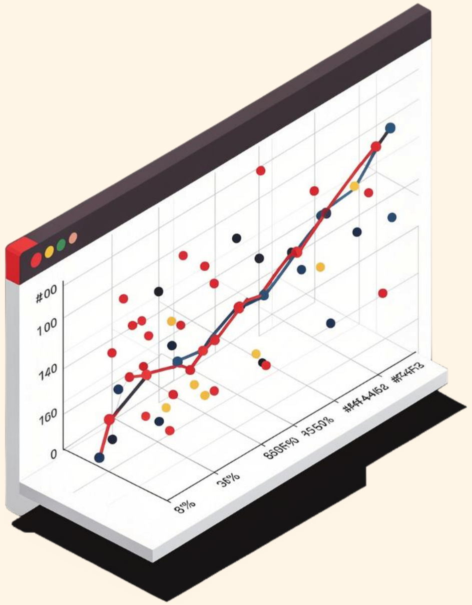
A supervised learning technique that predicts a **continuous** output variable from one or more input features

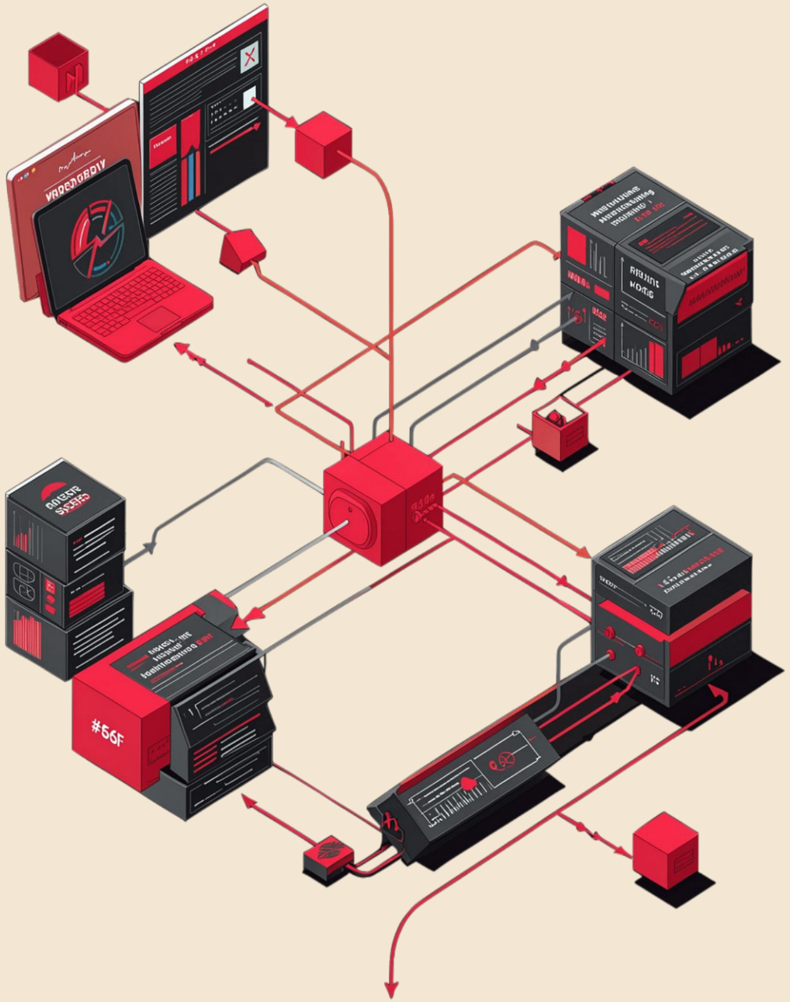
Regression vs. Classification

Regression → continuous values (e.g., **$350,000**)
Classification → discrete labels (e.g., **"spam / not spam"**)

Real-World Examples

- House price prediction
- Salary estimation
- Sales forecasting
- Stock price trends





How Regression Works

1

Input Features (X)

Independent variables fed into the model

2

Trained Model

Learns the best-fit relationship by minimizing prediction errors

3

Prediction (Y)

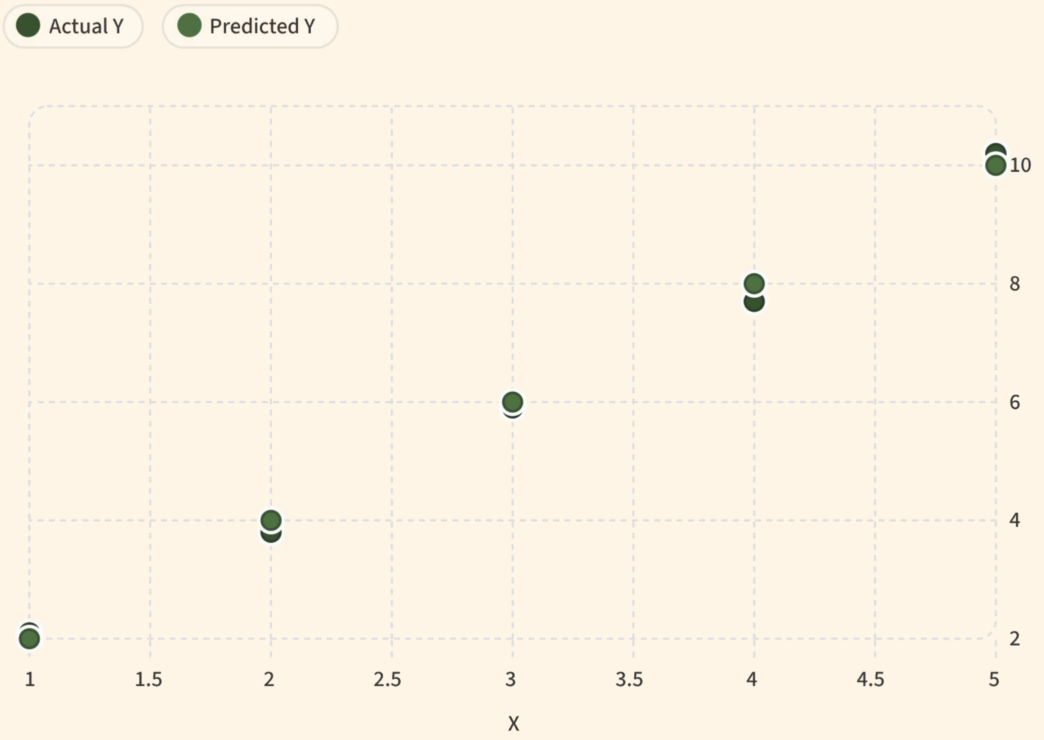
Estimated continuous output for any new X value

The model draws a **best-fit line** through training data — new inputs travel through the learned function to produce estimated outputs.

Types of Regression

Type	Complexity	Best Use Case	Key Advantage
Linear	Low	Simple linear trends	Interpretable
Multiple Linear	Low–Med	Multiple predictors	Scalable
Polynomial	Medium	Curved relationships	Flexible fit
Ridge	Medium	Multicollinearity	Shrinks coefficients
Lasso	Medium	Feature selection	Sparse model
Elastic Net	Med–High	High-dim data	Balanced penalty

Linear Regression Deep Dive



Actual Y vs. Predicted Y — vertical gaps represent **residuals** (prediction errors).

The Linear Equation

> $Y = \beta_0 + \beta_1 X + \varepsilon$

- β_0 — Intercept: where the line crosses the Y-axis
- β_1 — Slope: rate of change of Y per unit X
- ε — Error term: irreducible noise in the data

Least Squares Method

The model finds the line that **minimizes the sum of squared residuals** — the vertical distances between actual data points and the predicted line.

Assumptions of Linear Regression

Linearity

The relationship between X and Y must be linear

Independence

Observations must be independent of each other

Homoscedasticity

Residuals should have constant variance across all X values

Normality

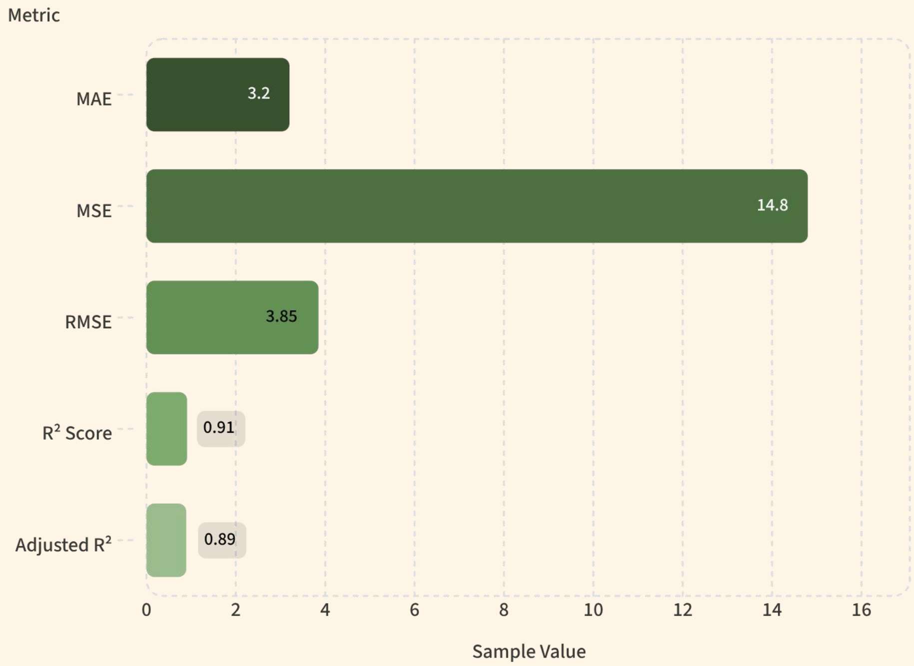
Residuals should follow a normal distribution

No Multicollinearity

Input features must not be highly correlated with each other

Violating these assumptions doesn't always break the model — but it can lead to biased estimates or unreliable predictions.

Model Evaluation Metrics



MAE — Mean Absolute Error

Average absolute difference. Easy to interpret, robust to outliers.

MSE / RMSE

Penalizes large errors heavily. RMSE is in the same unit as Y.

R^2 Score

Proportion of variance explained. **1.0 = perfect fit.**

Adjusted R^2

Penalizes unnecessary features — use when comparing models with different predictor counts.

Real-World Applications

Finance

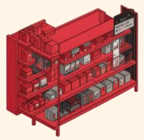
Stock price forecasting, credit risk scoring

Real Estate

House price prediction based on location, size, amenities

Healthcare

Predicting recovery time, drug dosage optimization



Retail & Energy

Sales forecasting, demand planning, energy consumption prediction

Climate

Weather forecasting, temperature trend modeling

Advantages, Limitations & Challenges

Performance Overview

Aspect	Detail
Interpretability	Coefficients directly show feature impact
Speed	Trains in milliseconds on most datasets
Outlier Sensitivity	A few extreme values can skew the line significantly
Nonlinearity	Polynomial/regularized variants needed for curves
Overfitting	Use Ridge/Lasso or cross-validation to control

✓ Advantages

- Interpretable coefficients
- Computationally fast
- Strong baseline model
- Works well on linear data

⚠ Limitations

- Sensitive to outliers
- Assumption-dependent
- Struggles with complex nonlinear patterns

🔧 Challenges

- Overfitting vs. underfitting trade-off
- Multicollinearity risks
- Careful feature selection required

Summary & Key Takeaways

01

Regression predicts continuous values

Uses learned relationships between input features and a target variable

02

Types range from simple to advanced

Linear → Multiple Linear → Polynomial → Ridge, Lasso, Elastic Net

03

Evaluate with the right metrics

Use MAE, RMSE, and R^2 — choose based on your specific use case

04

Applications are everywhere

Finance, healthcare, retail, climate, and far beyond

> "Regression is one of the most fundamental supervised learning techniques that helps predict continuous values and forms the foundation for many advanced machine learning algorithms."